Patau's syndrome. In Turner's syndrome the sex chromosome is XO and in Klinefelter's syndrome the condition is XXY. An idiogram refers to a diagrammatic representation of chromosomes.

Evaluation

1. Haemophilia is more common in males because it is a
   a) Recessive character carried by Y-chromosome
   b) Dominant character carried by Y-chromosome
   c) Dominant trait carried by X-chromosome
   d) **Recessive trait carried by X-chromosome**

2. ABO blood group in man is controlled by
   a) Multiple alleles
   b) Lethal genes
   c) Sex linked genes
   d) Y-linked genes

3. Three children of a family have blood groups A, AB and B. What could be the genotypes of their parents?
   a) $I^A I^B$ and $I^o I^o$ b) **$I^A I^o$ and $I^B I^o$**
   c) $I^B I^B$ and $I^A I^A$ d) $I^A I^A$ and $I^o I^o$

4. Which of the following is not correct?
   a) Three or more alleles of a trait in the population are called multiple alleles.
   b) A normal gene undergoes mutations to form many alleles.
   c) **Multiple alleles map at different loci of a chromosome.**
   d) A diploid organism has only two alleles out of many in the population.

5. Which of the following phenotypes in the progeny are possible from the parental combination AxB?
   a) A and B only
   b) A,B and AB only
   c) AB only
   d) **A,B,AB and O**

6. Which of the following phenotypes is not possible in the progeny of the parental genotypic combination $I^A I^O$ x $I^A I^B$?
   a) AB **b) O**
   c) A d) B

7. Which of the following is true about Rh factor in the offspring of a parental combination DdxDd (both Rh positive)?
   a) All will be Rh positive
   b) Half will be Rh positive
   c) About ¾ will be Rh negative
   d) About one fourth will be Rh negative

8. What can be the blood group of offspring when both parents have AB blood group?
   a) AB only **b) A, B and AB**
   c) A, B, AB and O d) A and B only

9. If the childs blood group is 'O' and fathers blood group is 'A' and mother's blood group is 'B' the genotype of the parents will be
   a) $I^A I^A$ and $I^B I^o$ b) **$I^A I^o$ and $I^B I^o$**
   c) $I^A I^o$ and $I^o I^o$ d) $I^o I^o$ and $I^B I^B$

10. XO type of sex determination and XY type of sex determination are examples of
    a) **Male heterogamety**
    b) Female heterogamety
    c) Male homogamety
    d) Both (b) and (c)

11. In an accident there is great loss of blood and there is no time to analyse the blood group which blood can be safely transferred?
    a) O and Rh negative
    b) O and Rh positive
    c)B and Rh negative
    d) AB and Rh positive

12. Father of a child is colourblind and mother is carrier for colourblindness, the probability of the child being colourblind is
    a) 25% **b) 50%**
    c) 100% d) 75%

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